

Grade 8
Unit 14
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

As part of a game, two fair six-sided dice are rolled.

- If the player rolls the same number on both dice, the player wins.
- In all other outcomes, the player loses.

1. What is the probability of winning the game?

$$\frac{6}{36} \text{ or } \frac{1}{6}$$

2. If the game is played 252 times, what is the predicted number of wins?

$$\frac{1}{6} \cdot 252 = 42 \text{ predicted wins}$$

3. If the winning prize is \$15 and it costs \$0.50 to play the game, would you want to play this game. Explain.

If you play 6 games just to win once, it would cost \$3 to play those six games and then you'd win \$15, so you'd end up making \$12. I'd want to play this game.

Shania is buying four strawberry plants for her garden. She is hoping to have six pounds of strawberries by the end of the season to make jam. When she reads more about the strawberry plants, she notices that the predicted germination rate is 65%. Round to the nearest thousandth.

4. Find the theoretical probability that only one of the plants will germinate.

$$\begin{aligned} P(\text{exactly one plant germinates}) &= P(\text{germ}) = 0.65 \\ &P(\text{not germ}) = 0.35 \end{aligned}$$

$$0.65 \cdot 0.35 \cdot 0.35 \cdot 0.35 \approx 0.028$$

5. Find the theoretical probability that half of the plants will germinate.

$$P(\text{half of the plants germinate}) =$$

$$0.65 \cdot 0.65 \cdot 0.35 \cdot 0.35 = 0.052$$

6. How likely is it that all four of Shania's strawberry plants germinate?

$$P(\text{all four plants germinate}) =$$

$$(0.65)^4 = 0.179$$

7. If she wants to have four strawberry plants that will germinate, explain how many plants she should buy.

$$0.65x = 4 \qquad x = 6 \text{ plants}$$

8. Dion rolls a fair eight-sided die 94 times. He correctly predicts that the die rolled an even number 47 times. Use the theoretical probability that the fair eight-sided die lands on an even number to explain how Dion made his prediction.

8-sided die has 4 even numbers, so the $P(\text{even number})$ is $\frac{4}{8}$ or $\frac{1}{2}$. So if he rolls the die 94 times, multiplied by $P(\text{even number})$, you'd get half of 94, which is 47.

Sasha has a standard 52-card deck of cards.

9. If she chooses three cards, with replacement, what is the probability that she chooses a jack, a queen, and a king of any suit in that order?

$$P(\text{Jack}) = \frac{4}{52} = \frac{1}{13} \quad P(\text{Queen}) = \frac{1}{13} \quad P(\text{King}) = \frac{1}{13}$$

$$\frac{1}{13} \cdot \frac{1}{13} \cdot \frac{1}{13} = \frac{1}{2,197}$$

10. If she chooses a card 143 times, how many outcomes including a face card (a jack, queen, or king) of any suit are expected?

$$P(\text{face card}) = \frac{12}{52} = \frac{3}{13}$$

$$\frac{3}{13} \cdot 143 = 33 \text{ outcomes}$$